

Training-Free Geometric Stabilization for Monocular 3D Detection in Advanced Rider-Assistance Systems

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Abstract

Monocular 3D object detectors fail on dynamic platforms like motorcycles, where roll and pitch variations violate the stable-camera assumption learned from automotive datasets. This geometric domain shift, which 2D augmentations cannot model, causes catastrophic 3D localization errors. We introduce Lean-Aware Geometric Stabilization (LGS), a training-free pre-processing module that corrects this instability at inference time. LGS applies a physically-grounded inverse reprojection to warp the input image to a canonical, stable orientation before it reaches a frozen, off-the-shelf detector. On a challenging motorcycle dataset, LGS recovers BEV AP@0.2 from 0.06 to 0.20 under a 10° roll drift, nearly matching the original 0.22 performance on stable views. The module adds only 5-10 ms of CPU latency, offering a practical method to deploy existing 3D detectors on geometrically unstable platforms without retraining.

Keywords: Monocular 3D Object Detection, Geometric Robustness, Viewpoint Augmentation, Training-Free Adaptation, Advanced Rider-Assistance Systems (ARAS).

1 Introduction

Monocular 3D object detectors [Liu et al. (2020), Zhang et al. (2021), Ma et al. (2021)] learn strong geometric priors from large-scale automotive datasets. This success is predicated on a stable camera platform that maintains a fixed extrinsic pose and a level horizon.

This "automotive assumption" is fundamentally violated on dynamic platforms like motorcycles, which exhibit continuous roll and pitch. During dynamic maneuvering (e.g., leaning in turns), vehicles exhibit continuous roll and pitch variations—violating the level-horizon extrinsic baseline. We term this Dynamic Extrinsic Drift (DED), which induces geometric domain shift in camera projection while visual appearances remain identical. Consequently, a detector trained on stable data (e.g., KITTI [Geiger et al. (2012)]) fails catastrophically when its learned priors are invalidated, as shown in Figure 1. Standard 2D training augmentations (e.g., planar rotations) fail because they treat pitch/roll as affine image shifts, ignoring non-linear perspective distortions

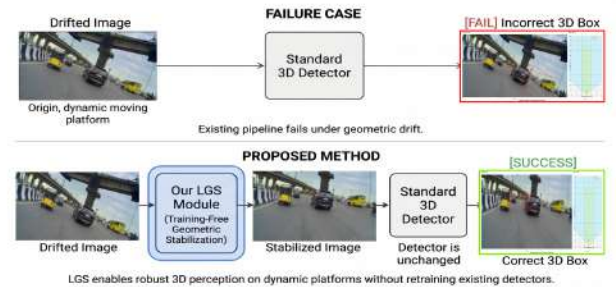


Figure 1: **Core Idea:** (Top) A standard detector fails on a geometrically drifted image. (Bottom) Our training-free LGS module stabilizes the input, enabling the same detector to succeed without retraining. and violating depth projection physics. This work

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therefore addresses whether a lightweight, training-free geometric stabilization can counteract this domain shift, enabling a pre-trained detector to generalize to dynamic motion without costly retraining.

We introduce *Lean-Aware Geometric Stabilization (LGS)*, a model-agnostic pre-processing module that corrects for extrinsic drift at inference time. Our contributions are:

1. We formalize DED, a physically-grounded viewpoint augmentation, to create a challenging benchmark for evaluating geometric robustness.
2. We propose LGS, a model-agnostic, training-free pre-processing module that restores geometric consistency to input images at inference time.
3. We demonstrate empirically that LGS enables a detector trained only on stable data to generalize to out-of-distribution camera poses, substantially recovering 3D localization accuracy.

2 Prior Art Study

Our work is positioned at the intersection of geometric robustness, test-time adaptation, and perception for dynamic platforms. **Geometric Robustness.** The fragility of monocular 3D detectors to camera pose variations is well-documented [Moon et al. (2023)]. A common defense is training-time augmentation; for instance, GRAMO [Guan et al. (2024)] applies ground-plane constraint resampling to synthesize geometrically consistent samples during training. Our DED simulation can serve this purpose, but our primary contribution, LGS, is a complementary, inference-time stabilization technique that robustifies existing detectors without retraining. This is distinct from work on camera *intrinsic* robustness [Zhang et al. (2026)], as we focus on *extrinsic* drift. **Test-Time Adaptation (TTA).** Another strategy is to adapt the model at test time. MonoTTA [Lin et al. (2024)] uses self-supervision to adapt to domain shifts, while other methods rectify latent features using online pose estimates [Zhou et al. (2021)]. These approaches often require access to model internals or add significant overhead. In contrast, LGS is a model-agnostic pre-processing step. By correcting geometry in the image space *before* the detector, it provides a simpler, modular, and training-free solution. It is a form of test-time adaptation that operates on the input data, not the model, and is distinct from methods correcting rolling shutter artifacts [Mo et al. (2022)], which address intra-frame distortions. **Attitude Estimation and Hardware Compensation.** LGS assumes vehicle roll and pitch are available from an IMU. The problem of estimating these angles [Ok et al. (2021)] is orthogonal to our work; we focus on how to use this information for visual stabilization. Similarly, our approach is distinct from hardware-level compensation like Optical Image Stabilization (OIS) [Liu et al. (2022)], which handles high-frequency camera shake. LGS addresses a different problem: large-scale, low-frequency changes in the vehicle’s extrinsic pose (e.g., a motorcycle lean), a challenge not addressed by OIS.

3 Methodology: Geometric Augmentation and Stabilization

Our framework is built on the principle of virtual camera reprojection. We first define the core geometric transformation before detailing its dual use for simulating drift (DED) and correcting it (LGS).

3.1 Geometric Formulation

The core of our method is a physically-grounded warp that maps pixels from a source camera view to a virtual, rotated one. Assuming two cameras share an optical center and intrinsic matrix $\mathbf{K}$, but are rotated by $\mathbf{R}_{\text{drift}}$ relative to each other, we can find the location $\mathbf{x}_{\text{src}}$ in the source image corresponding to a pixel $\mathbf{x}_{\text{dst}}$ in the destination (virtual) image via inverse reprojection:

$$\mathbf{x}_{\text{src}} \sim \mathbf{K} \mathbf{R}_{\text{drift}}^{-1} \mathbf{K}^{-1} \mathbf{x}_{\text{dst}} \quad (1)$$

This equation traces a ray from the virtual camera’s image plane, rotates it into the source camera’s frame, and re-projects it. The drift rotation $\mathbf{R}_{\text{drift}}$ models the orientation change, which we parameterize by roll (θ_r) and pitch (θ_p) angles. The rotation is constructed in the camera’s local coordinate system (X-right, Y-down, Z-forward):

$$\mathbf{R}_{\text{drift}} = \mathbf{R}_z(\theta_r)\mathbf{R}_x(\theta_p) \quad (2)$$

This formulation is powerful because it is agnostic to any global ‘world’ coordinate system used for 3D annotations. It operates purely on camera geometry, making it adaptable to diverse dataset conventions.

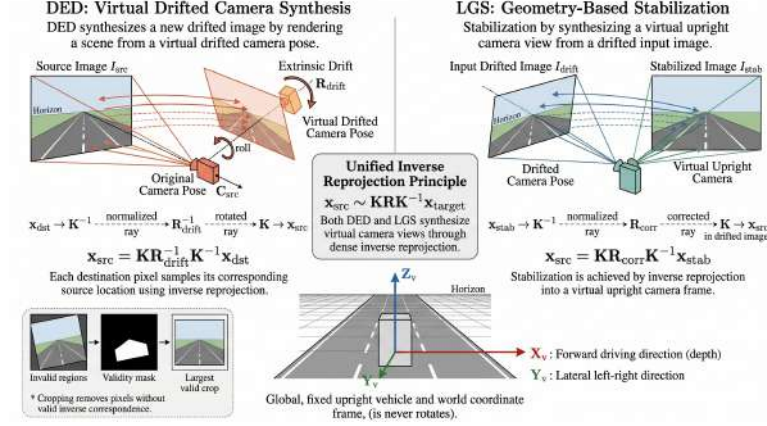


Figure 2: **Geometric Principle of DED and LGS:** Our framework is based on inverse reprojection. (Left) To simulate drift (DED), we trace pixels from a virtual drifted camera back to the original source image using the inverse drift rotation ($\mathbf{R}_{\text{drift}}^{-1}$). (Right) To correct drift (LGS), we trace pixels from a virtual upright camera back to the drifted input image using the corrective rotation ($\mathbf{R}_{\text{corr}}$). This visually explains the geometry-consistent warp defined in Eq. 1.

3.2 Dynamic Extrinsic Drift (DED) Simulation

To evaluate model robustness, we use Eq. 1 to simulate the effect of drift. This process, which we call Dynamic Extrinsic Drift (DED), generates a new, geometrically consistent image from the perspective of a virtual, drifted camera. The resulting map is used with bilinear interpolation to produce the warped image. To remove boundary artifacts, the warped image is cropped to the largest axis-aligned inner rectangle using corner ray-casting; while this slightly reduces peripheral FOV, key central drivable regions remain preserved.

Crucially, DED is a full 3D augmentation that transforms 3D ground truth labels in addition to the image, a key differentiator from 2D-only methods. The exact transformation depends on the annotation coordinate system. For datasets with a stable world frame (e.g., our ARAS dataset), the drift $\mathbf{R}_{\text{drift}}$ updates the camera’s extrinsic rotation matrix. For datasets where annotations are in the camera’s coordinate system (e.g., KITTI), the 3D ground truth points are simply rotated by $\mathbf{R}_{\text{drift}}$. This rigorous label transformation (Figures 3, 4) is essential for correct model evaluation.



Figure 3: **DED on our ARAS Dataset.** (Top) Original image and 3D box. (Middle) Warped images for different drift parameters. (Bottom) Final cropped images. Our method applies the same 3D rotation to the ground truth, producing new, physically-correct labels for the augmented views. Drifts (L-R): (R:+20°, P:+10°), (R:-20°, P:+5°), (R:-10°, P:-5°).



Figure 4: **DED Generality on KITTI.** We apply DED to KITTI to demonstrate the generality of our augmentation. (Top) Original image. (Middle) Warped images showing perspective distortion. (Bottom) Final cropped images with transformed ground truth. This creates a challenging benchmark for geometric robustness on standard automotive data.



Figure 5: **DED on nuScenes.** To further prove generality, we apply DED to the nuScenes dataset [Caesar et al. (2020)], which also uses a camera-centric coordinate system. This demonstrates DED’s utility as a cross-dataset tool for robustness analysis. Drifts (L-R): (R:+15°, P:+8°), (R:0°, P:+8°), (R:-15°, P:-8°).

3.3 Lean-Aware Geometric Stabilization (LGS)

LGS is the corrective counterpart to DED, applied at inference time. Given a drifted input image and an estimate of the drift angles (e.g., from an IMU), LGS computes a corrective rotation $\mathbf{R}_{\text{corr}} = \mathbf{R}_{\text{drift}}^{-1}$. It then applies the same inverse reprojection warp from Eq. 1 to transform the image back to a stable, upright orientation that matches the detector’s training distribution. This stabilized image is then passed to the off-the-shelf detector, as shown in Figure 1. The entire process is a single, efficient pre-processing step.

4 Experiments

Our experiments are designed to quantify the performance degradation caused by DED and measure the recovery provided by LGS.

4.1 Setup

Datasets and Protocol. We use a challenging in-house **ARAS dataset** from a motorcycle and the standard **KITTI benchmark** [Geiger et al. (2012)]. To mirror real-world lean dynamics captured in dataset telemetry, we fine-tune a SMOKE detector [Liu et al. (2020)] on stable ARAS subsets (roll $< 5^\circ$) and test on simulated DED sequences matching real-world lean trajectories. This specialist model is then evaluated on its ability to generalize to images with significant, unseen drift (up to 25°), as shown in Figure 6. We evaluate three scenarios: (1) **Baseline** (on original images), (2) **Drifted** (on DED-warped images), and (3) **Stabilized** (Drifted images corrected by LGS). We report BEV AP@0.2 for 3D localization and Yaw Variance for orientation stability. For ablation, we compare against a naive 2D rotation correction. We also show DED’s generality on nuScenes [Caesar et al. (2020)] (Figs. 4, 5).

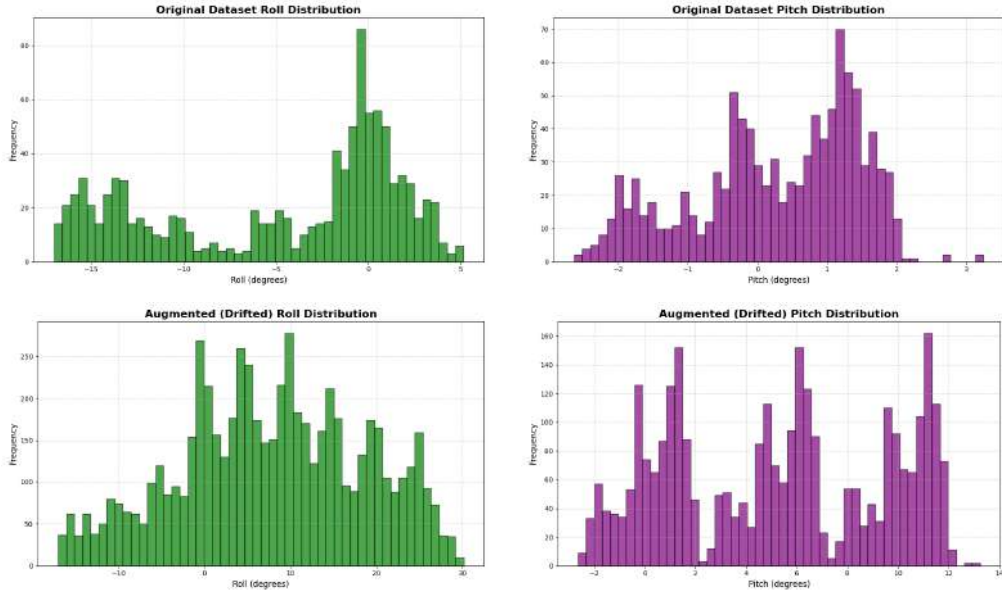


Figure 6: **Geometric Domain Shift for Evaluation.** The training set distributions for roll and pitch (top row) contain only stable views near 0° . In contrast, our DED-augmented test set (bottom row) covers a wide range of angles, creating a challenging OOD scenario to test generalization.

4.2 Results and Analysis

Quantitative Results. As shown in Table 1, the performance of the baseline model, trained on upright images, degrades severely under unseen drift. At 10° roll, BEV AP@0.2 plummets from 0.224 to

0.056, confirming the model’s failure to generalize geometrically. LGS provides a substantial recovery, restoring performance to 0.199 (+0.143 AP), nearly reaching the original performance on stable views and demonstrating its effectiveness at bridging the geometric domain gap.

Table 1: Performance of the baseline model, trained on stable images, degrades catastrophically under roll-induced DED. LGS substantially recovers performance, enabling generalization to unseen drift angles at inference time. Best results on drifted data are in bold.

Roll Drift	Method	BEV AP@0.2 $\uparrow$	Yaw Variance (rad ²) $\downarrow$
0°	Baseline	0.224	0.01
10°	Drifted	0.056	0.35
	LGS (Ours)	0.199 ($\Delta+0.143$)	0.04 ($\Delta-0.31$)

4.3 Ablation Study

To prove our 3D reprojection necessity, we compare LGS against naive 2D correction (Table 2), which simply applies inverse 2D affine image rotation without accounting for intrinsic focal length K or 3D perspective rays. The results are unequivocal: simple 2D rotation is geometrically insufficient and degrades performance further, confirming that correctly modeling the 3D-to-2D projection is essential.

Table 2: Ablation study at 10° roll drift. The naive 2D correction is insufficient, proving the necessity of our physically-grounded geometric reprojection.

Method (at 10° roll)	BEV AP@0.2 $\uparrow$	Yaw Variance (rad ²) $\downarrow$
Drifted (No Correction)	0.056	0.35
+ Naive 2D Un-rotation	0.040	0.75
+ LGS (Ours)	0.199	0.04

4.4 Qualitative Analysis

Figure 7 provides a visual comparison. On drifted images, the detector is confused by distorted geometric cues, leading to catastrophic failures. After our LGS module stabilizes the input, the same detector produces a precise 3D bounding box, directly illustrating how LGS bridges the geometric domain gap.

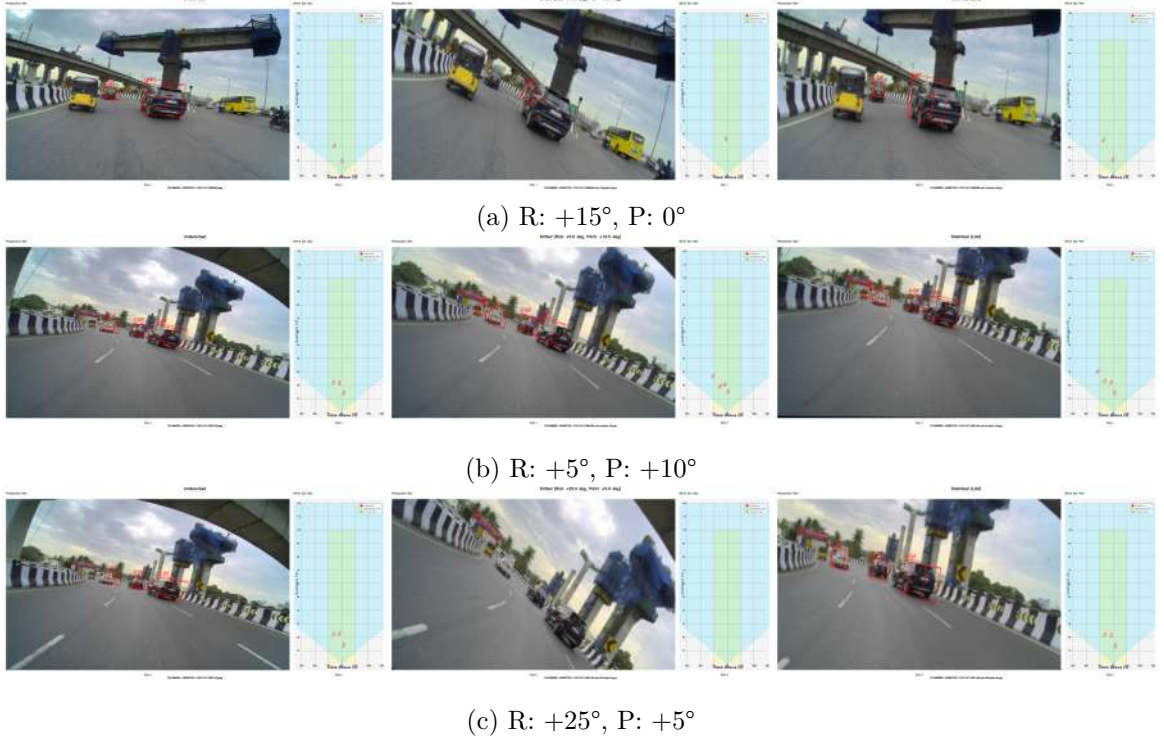


Figure 7: Qualitative Analysis: Each row shows a test case with its drift angles (R: Roll, P: Pitch). Left: Baseline prediction on original image. Middle: Catastrophic failure on DED-warped image. Right: Recovered, accurate prediction after applying our LGS module. LGS enables robust detection under significant, unseen drift.

4.5 Runtime Analysis

LGS is computationally efficient. Implemented in OpenCV on a CPU, the warp adds only **5-10 ms** of latency. With a detector inference time of 39.9 ms on a Tesla T4 GPU, the total pipeline latency is **45-50 ms** per frame (>20 FPS). This minimal overhead makes LGS practical for real-time deployment. Sensitivity tests show LGS maintains robust detection under typical IMU noise ($\pm 1^\circ$ angle jitter) and temporal lag (< 20 ms sensor delay).

5 Conclusion

We have demonstrated that dynamic camera extrinsics on platforms like motorcycles cause a critical geometric domain shift that cripples standard monocular 3D detectors. To solve this, we introduced Lean-Aware Geometric Stabilization (LGS), a lightweight, training-free pre-processing module that restores geometric consistency at inference time. By applying a physically-grounded inverse reprojection, LGS enables a frozen, off-the-shelf detector to generalize to significant, unseen camera poses, substantially recovering 3D localization performance. This work not only provides an immediate, practical solution for deploying existing detectors on dynamic platforms but also opens future avenues. These include using our DED simulation as a training-time augmentation to build inherent robustness into models and extending LGS to handle extreme angles and sensor noise. Ultimately, our framework underscores the primacy of geometric integrity and offers an efficient path toward robust 3D perception in challenging, real-world environments.

References

- [Liu et al. (2020)] Liu, Z., Wu, Z., & Tóth, R. (2020). "SMOKE: Single-Stage Monocular 3D Object Detector via Keypoint Estimation." *CVPR Workshops*, pp. 4289-4298.
- [Zhang et al. (2021)] Zhang, Y., Lu, J., & Zhou, J. (2021). "Objects Are Different: Flexible Monocular 3D Object Detection." *CVPR*, pp. 3289-3298.
- [Ma et al. (2021)] Ma, X., et al. (2021). "Delving into Localization Errors for Monocular 3D Object Detection." *CVPR*, pp. 339-348.
- [Geiger et al. (2012)] Geiger, A., Lenz, P., & Urtasun, R. (2012). "Are we ready for Autonomous Driving? The KITTI Vision Benchmark Suite." *CVPR*, pp. 3354-3361.
- [Guan et al. (2024)] Guan, H., Song, C. & Zhang, Z. (2024). "GRAMO: geometric resampling augmentation for monocular 3D object detection." *Frontiers of Computer Science*, 18, 185706. <https://doi.org/10.1007/s11704-023-3242-2>
- [Lin et al. (2024)] Lin, H., Zhang, Y., Niu, S., Cui, S., & Li, Z. (2024). "Fully Test-Time Adaptation for Monocular 3D Object Detection." In *European Conference on Computer Vision*.
- [Zhou et al. (2021)] Zhou, Y., He, Y., Zhu, H., Wang, C., Li, H., & Jiang, Q. (2021). "Monocular 3D Object Detection: An Extrinsic Parameter Free Approach." *arXiv preprint arXiv:2106.15796*.
- [Moon et al. (2023)] Moon, S., Bae, J., & Im, S. (2023). "Rotation Matters: Generalized Monocular 3D Object Detection for Various Camera Systems." *arXiv preprint arXiv:2310.05366*.
- [Zhang et al. (2026)] Zhang, Z., Kumar, A., Liu, X. (2026). "Towards Intrinsic-Aware Monocular 3D Object Detection." *arXiv preprint arXiv:2603.27059*.
- [Chen et al. (2021)] Chen, H., Huang, Y., Tian, W., Gao, Z., & Xiong, L. (2021). "MonoRUn: Monocular 3D Object Detection by Reconstruction and Uncertainty Propagation." In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10374-10383.
- [Caesar et al. (2020)] Caesar, H., et al. (2020). "nuScenes: A Multimodal Dataset for Autonomous Driving." In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11618-11628.
- [Mo et al. (2022)] Mo, J., Islam, M.J. & Sattar, J. (2022). "IMU-Assisted Learning of Single-View Rolling Shutter Correction." In *Proceedings of the 5th Conference on Robot Learning*, in *Proceedings of Machine Learning Research*, vol. 164, pp. 861-870. Available from <https://proceedings.mlr.press/v164/mo22a.html>.
- [Ok et al. (2021)] Ok, M., Ok, S., & Park, J. H. (2021). "Estimation of Vehicle Attitude, Acceleration, and Angular Velocity Using Convolutional Neural Network and Dual Extended Kalman Filter." *Sensors*, 21(4), 1282. <https://doi.org/10.3390/s21041282>
- [Liu et al. (2022)] S. Liu, H. Li, Z. Wang, J. Wang, S. Zhu and B. Zeng, "DeepOIS: Gyroscope-Guided Deep Optical Image Stabilizer Compensation," in *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 32, no. 5, pp. 2856-2867, May 2022, doi: 10.1109/TCSVT.2021.3103281.